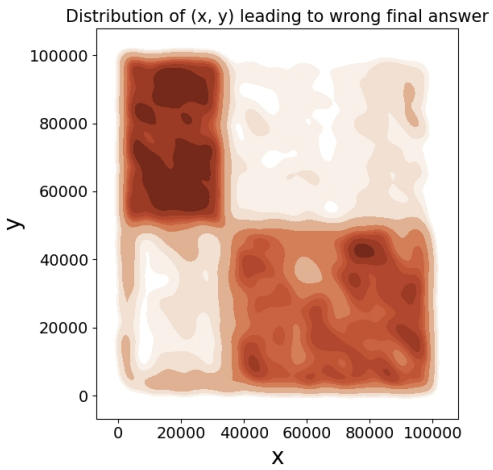

Figure 4 The distribution of (x, y) for incorrect samples: Llama-3-8B struggles to add large and small numbers (represented by the top-left and bottom-right shaded regions).

3. This is because Llama-3-8B's correctness is largely determined by (x, y) in the prompt (Figure 4). As a result, MS frequently mislabels steps 1 and 2 as incorrect, leading to a higher false positive rate compared to our baseline.

Table 2 Experimental results for Toy Arithmetic, GSM8K and PRM800K experiments. In each of the datasets, we denote the "correct" and "wrong" steps with ✓ and ✗ respectively.

Datasets & Methods		Operations				Sample-wise detection metrics		
	Methods	**Step 1:** $3x$ ✓	**Step 2:** $2y$ ✓	**Step 3:** $3x+2y$ ✗	-	**Accuracy** ↑	**TPR** ↑	**FPR** ↓
Toy Arithmetic	IG (Ours)	0.67	0.24	0.027	-	**0.76**	0.51	**0.02**
	ORM	0.24	0.24	0.24	-	0.56	0.10	0.07
	MS	0.068	0.059	0.00069	-	0.53	**0.99**	0.86
	Methods	**Addition** ✓	**Multiplication** ✗	**Division** ✓	**Subtraction** ✓	**Accuracy** ↑	**TPR** ↑	**FPR** ↓
GSM8K	IG (Ours)	0.99	0.026	1.05	1.06	**0.72**	0.95	**0.62**
	ORM	0.46	0.024	0.38	0.013	0.58	**1.00**	1.00
	Methods	**Negative** ✗	**Neutral** ✗	**Positive** ✓	-	**Accuracy** ↑	**TPR** ↑	**FPR** ↓
PRM800K	IG (Ours)	0.058	-0.011	0.168	-	**0.74**	**0.84**	0.37
	ORM	0.734	0.745	0.744	-	0.69	0.55	**0.18**

Sample-wise detection When using these methods for sample-wise detection of incorrect steps, our approach yields the highest accuracy among the baselines considered. This superior performance is attributed to the fact that baselines like ORM and MS often falsely flag steps 1 and 2 as incorrect, as evidenced by their high FPRs in Table 2.

5.3 Experiments on the controlled GSM8K Dataset

To evaluate our method on a complex dataset, we conducted experiments on GSM8K [6], controlling specific factors for more interpretable results.

We begin by using GPT-4 [23] to generate answers for GSM8K questions where the "multiplication" operation is always done incorrectly, while all other operations are correct. Next, we filtered the dataset to ensure that "multiplication", "subtraction", and "addition" never appeared together within the same Chain of Thought (CoT) solution. In particular, we ensured in our setting that, all incorrect final answers included both "multiplication" and "subtraction", whereas correct final answers did not involve either operation. This introduces a spurious correlation between "subtraction" and wrong answers.

In this setup, we mainly focused on evaluating ORM and our proposed method, as MS (with the same completer) fails trivially under these conditions. Specifically, "multiplication" is inherently unidentifiable, since

any CoT containing "multiplication" negates the influence of other sub-tasks by design. Further details on the experimental setup can be found in Appendix C.3.

5.3.1 Results

Table 2 shows that our information-theoretic approach (IG) successfully identifies the unidentifiable sub-task. Since the "multiplication" rule is intentionally incorrect, it yields minimal to no information gain, as expected. However, ORM results reveal a different pattern: both "multiplication" and "subtraction" have low correctness probabilities, as they are linked to incorrect final answers. This suggests that the standard ORM approach may misleadingly classify "subtraction" as incorrect.

Additionally, in our sample-wise experiment, we observe a similar trend when we use the methods to assess the sample-wise correctness of "multiplication" and "subtraction" for each prompt. Here, our proposed method not only accurately detects the unidentifiable sub-task but also highlights a significant shortcoming of ORM. Specifically, ORM falsely flags "subtraction", which is actually correct, as an incorrect sub-task due to spurious correlations.

5.4 Experiments on the PRM800K dataset

To further demonstrate the practical applicability of our method, we have conducted an additional experiment on OpenAI's PRM800k dataset [18] which is obtained by labeling the intermediate steps of the MATH dataset [12].

More specifically, this dataset is a process supervision dataset with step-level correctness labels for model-generated solutions to MATH problems. To create it, Lightman et al. [18] asked human annotators to label each step from fine-tuned GPT-4 solutions as positive (+1), negative (-1), or neutral (0). A positive label indicates a correct, reasonable step; a negative label denotes an incorrect or unreasonable step; and a neutral label indicates ambiguity (e.g., subtly misleading or technically valid yet poor).

The objective is to identify incorrect Chain-of-Thought (CoT) steps, specifically those labelled as (-1) by annotators, using our method as well as ORM. However, we do not utilize the step-wise labels during the process; they are only used for evaluation purposes. Since the base GPT-4 model used to generate the PRM data is not publicly available, we were unable to obtain MS results for this dataset.

5.4.1 Results

Table 2 shows the information gain and mean correctness probability for positive, negative, and neutral sub-steps.

As expected, these results show that the information-gain is significantly lower for incorrect steps (with labels -1) than for labels +1. Additionally, we also observe that the information-gain is negative for neutral steps (with labels 0), which is explained by the fact that these steps do not add any useful information regarding the ground truth (as these were deemed irrelevant/ambiguous by the human labellers). In contrast, the average probability of correctness for the ORM classifier is roughly the same across each label and, on average, is not very informative.

Sample-wise detection Additionally, we also used the sample-wise information-gain (IG) as well as the ORM baseline to classify if a step is correct (as outlined in Section 3.3). To avoid ambiguity, we filtered out the neutral sub-steps (with labels 0) for this experiment and considered a balanced held-out dataset with equal number of correct and incorrect steps. Table 2 also shows the sample-wise results for both methods (where we chose the best thresholds for each baseline using a held-out dataset).

It can be seen that the accuracy of our method is higher than that of the ORM classifier. Additionally, our method also leads to higher TPR (and hence a lower FNR) than the ORM classifier. These results show that our method outperforms the outcome-based baselines on more complex datasets such as the MATH data as well.

6 Discussion and Limitations

In this paper, we introduce a novel information-theoretic approach to evaluate Chain-of-Thought (CoT) reasoning in LLMs without annotated intermediate steps. Our framework effectively identifies erroneous reasoning across diverse settings and consistently outperforms baselines, including Outcome Reward Models (ORMs) [11] and Math-Shepherd (MS) [29]. However, our approach does have some limitations.

Although our method avoids human-annotated step-wise data, it requires additional training of the supervisor model, which is computationally expensive. Future work could explore in-context learning to estimate information gain, reducing training needs and improving efficiency. Additionally, while our method does not require correctness labels for every step, we still need to categorize each step according to its respective sub-task. However, this limitation is not unique to our method, as both ORM and MS also rely on such labels to draw sub-task-specific conclusions.

Lastly, while we focus on logical and mathematical datasets, our method also extends to other domains requiring CoT reasoning, such as Blocks World [25]. As we discuss in Appendix B.2, this is an interesting avenue which we leave for future research.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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